

SUOMUSSALMI, Finland

WMO: 028790

Lat: 64.900N

Lon: 29.017E

Elev: 735

StdP: 14.31

Time Zone: 2.00 (EUE)

Period: 90-00

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-19.8	-14.4	-24.3	1.5	-19.8	-18.5	2.0	-14.2	18.5	27.4	16.4	26.9	0.7	150	0.674

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.3	74.9	60.8	71.8	59.9	68.8	58.5	63.6	70.8	61.6	68.5	59.8	66.6	5.4	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
60.7	81.5	67.4	58.5	75.3	64.3	56.7	70.5	62.8	29.3	71.1	27.9	68.8	26.6	66.8	70.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.1	13.8	12.0	DB	-26.5	78.8	4.8	2.8	-30.0	80.7	-32.8	82.4	-35.4	83.9	-38.9	85.9	(4)
(5)			WB	-26.6	65.8	4.7	1.9	-30.0	67.2	-32.7	68.2	-35.4	69.3	-38.8	70.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	33.7	13.3	14.1	22.0	30.9	41.5	53.7	57.7	53.9	44.1	33.7	21.3	16.6
	DBStd	18.27	12.41	13.82	8.65	8.97	7.74	7.31	5.26	5.61	6.32	9.33	10.62	12.68
	HDD50	6543	1138	1004	868	573	288	42	6	25	198	505	862	1034
	HDD65	11441	1603	1424	1333	1023	730	345	232	344	627	969	1312	1499
	CDD50	590	0	0	0	0	24	154	244	147	21	0	0	0
	CDD65	13	0	0	0	0	0	7	5	1	0	0	0	0
	CDH74	126	0	0	0	0	5	78	29	14	0	0	0	0
	CDH80	9	0	0	0	0	0	8	1	0	0	0	0	0
Wind	WSAvg	5.0	4.7	4.8	5.7	5.2	5.2	5.2	5.1	4.7	4.9	5.3	4.9	4.7
Precipitation	PrecAvg	25.0	1.7	1.4	1.4	1.2	2.1	2.4	3.2	2.9	2.5	2.4	2.2	2.0
	PrecMax	30.5	2.8	2.6	2.0	2.2	4.0	5.6	4.6	5.9	4.4	5.4	4.4	3.7
	PrecMin	18.1	0.4	0.2	0.3	0.4	0.4	1.1	0.9	1.0	0.3	0.3	0.4	0.4
	PrecStd	3.1	0.6	0.7	0.5	0.6	0.9	1.0	0.9	1.3	1.0	1.2	1.1	0.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	37.4	39.0	41.5	56.4	74.0	81.0	77.6	76.6	67.3	52.9	42.0	36.6
		MCWB	34.7	35.1	36.8	45.5	56.2	62.9	62.9	62.7	58.6	50.7	41.2	34.5
	2%	DB	33.8	35.9	38.0	51.6	67.6	76.1	73.7	72.4	60.7	49.8	38.7	34.8
		MCWB	32.5	34.3	33.8	42.8	53.4	60.3	61.2	60.5	53.6	48.1	37.7	33.7
	5%	DB	31.7	33.2	35.5	47.8	61.6	71.5	71.1	69.0	56.7	48.1	36.6	33.4
		MCWB	30.9	32.2	32.3	40.4	50.2	59.0	60.3	59.6	52.0	46.7	35.7	32.6
	10%	DB	29.5	31.5	33.2	44.1	56.0	67.6	68.3	65.3	53.8	46.1	34.1	32.0
		MCWB	28.7	30.6	31.1	38.2	47.1	57.4	59.0	57.7	50.7	44.7	33.2	31.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	35.0	36.5	37.7	46.7	58.6	66.5	67.0	63.7	59.3	51.1	41.6	35.4
		MCDB	36.8	37.9	41.3	53.9	66.1	75.8	73.9	72.5	66.0	52.2	42.2	35.8
	2%	WB	32.9	34.3	34.4	43.6	54.5	63.5	63.8	62.2	55.5	48.6	37.8	33.9
		MCDB	33.7	35.2	36.6	50.2	66.9	70.4	70.5	70.1	59.2	49.7	38.4	34.6
	5%	WB	31.0	32.4	32.9	41.2	51.0	60.8	61.7	60.1	53.2	46.9	35.8	32.6
		MCDB	31.6	33.2	34.9	46.9	60.7	69.1	68.0	67.0	55.8	48.1	36.4	33.2
	10%	WB	28.6	30.5	31.4	38.8	47.7	58.1	60.1	58.1	50.7	44.7	33.2	31.1
		MCDB	29.4	31.5	33.1	43.1	54.9	66.5	66.1	64.5	53.4	45.8	34.0	31.8
Mean Daily Temperature Range	5% DB	MDBR	12.7	11.9	13.9	15.3	15.8	15.7	15.3	15.1	11.9	7.8	9.3	10.5
		MCDBR	11.2	7.4	14.5	17.9	26.0	22.4	21.0	22.7	16.3	8.1	7.1	8.8
		MCWBR	11.0	7.0	11.5	10.9	13.9	10.9	11.2	12.6	10.3	7.2	7.2	8.9
	5% WB	MCDBR	11.3	7.4	11.2	16.7	24.2	17.8	16.9	19.8	14.0	8.3	7.3	8.3
		MCWBR	11.3	7.3	9.5	10.4	13.4	10.6	9.6	11.7	9.7	7.6	7.4	8.4
Clear-Sky Solar Irradiance	taub	0.199	0.269	0.298	0.332	0.339	0.339	0.361	0.341	0.317	0.294	0.350	0.198	
	taud	1.933	2.113	2.163	2.258	2.463	2.488	2.488	2.444	2.482	2.533	2.387	2.657	1.957
	Ebn at Noon	100	195	245	265	276	278	267	263	246	193	95	29	
	Edh at Noon	8	19	28	33	30	30	30	26	20	15	7	3	
All-Sky Solar Radiation	RadAvg	39	199	652	1190	1475	1629	1530	1132	619	219	49	12	
	RadStd	4	41	59	121	154	115	156	112	65	33	10	1	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (22 neighbors)	+1.03	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page